

Ekaansh Patel

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EDUCATION

University of Michigan, College of Literature, Science, and The Arts

Expected May 2028

B.S. in Computer Science, GPA: 3.6

Ann Arbor, MI

Awards: Fall 2024 and Fall 2025 University Honors List

Coursework: Data Structures and Algorithms, Computer Science Pragmatics (Developer Tools and Ecosystems), Introduction to Computer Architecture, Foundations of Computer Science (Computational Complexity and Computability), Introduction to Machine Learning (Fall 2026), Web Systems (Fall 2026)

TECHNICAL SKILLS

Languages: C++, Python, HTML/CSS, TypeScript, JavaScript, Java

Libraries/Frameworks: Next.js, React, Tailwind CSS, Playwright, Pandas, NumPy

Developer Tools: VS Code, Git, Linux/Unix, npm, Vite

PROJECTS

Formula 1 Ultimate Team Game

- Engineered a full-stack, arcade-style drafting game using Next.js, TypeScript, and Tailwind CSS, simulating a 24-race Formula 1 season based on historical team and driver data.
- Designed a stateless architecture utilizing React state for local session memory and Next.js Server Actions to securely process game loops, reducing cloud database connection overhead.
- Implemented a secure backend validation and rating pipeline via Prisma and Neon PostgreSQL, preventing client-side score manipulation by calculating ultimate team success formulas entirely on the server.
- Architected data pipelines utilizing Python, Pandas, and NumPy to scrape, normalize, and inject historical sports metrics from the RacingHub API into relational database schemas.

Formula 1 Race Analytics Platform

- Engineered a predictive Formula 1 dashboard using React, JavaScript, Vite, and Tailwind CSS, to simulate and visualize race progressions based on predictive modeling.
- Developed a robust data processing pipeline in Python, utilizing Pandas and NumPy to parse and aggregate massive race datasets via the FastF1 and OpenF1 APIs.
- Designed a structured data integration layer to feed processed Python model outputs directly into the React frontend, optimizing state management for real-time interval metrics.

PROJECT TEAMS & ACTIVITIES

Wolverine Sports Analytics | *Project Officer, Analytics Member*

Sept 2025 – Present

- Appointed as Project Officer (Effective Spring 2026) to oversee development timelines, manage software deliverables, and coordinate project execution across the analytics team.
- Collected and scraped player statistics from multiple sports websites, normalizing data into unified CSV datasets to feed predictive pipelines.
- Implemented a fantasy football projection engine that estimates waiver-wire player point totals using a mathematical model incorporating player usage and historical performance.

Michigan Mars Rover Team (Mrover) | *Teleoperations Software Subteam*

September 2025 – Present

- Engineered responsive UI features for the online base station, implementing Playwright end-to-end test suites to validate functionality and prevent regressions.
- Enhanced the operator experience by redesigning the data interface, transitioning from legacy data tables to intuitive, clickable components mapped directly onto a visual model of the rover.

PROFESSIONAL EXPERIENCES

Arati Patel & Associates, PLC | *Office Administrator*

June 2026 – August 2026

- Managed high-volume client communications and streamlined the distribution, collection, and organization of sensitive tax documentation.
- Executed monthly bank reconciliations by aggregating and verifying financial data across multiple accounts, increasing administrative efficiency and ensuring accurate record-keeping.
- Executed the data cleansing and reorganization of legacy client archives for a major software migration, saving the team an estimated two weeks of manual transition time.